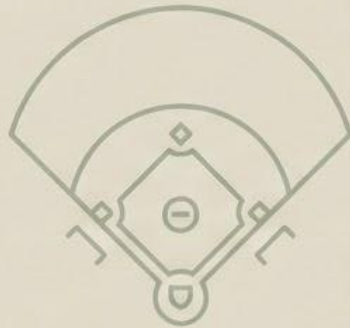


Predicting Pitcher ERA Using Advanced Metrics



INTRODUCTION

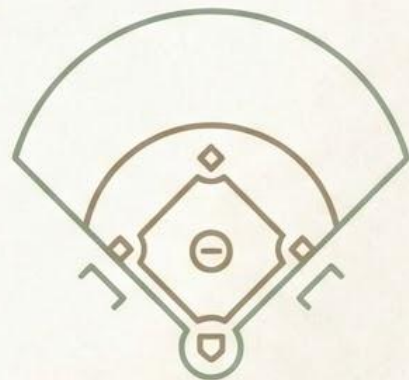
RESEARCH QUESTION: To what extent do underlying performance metrics explain variation in ERA among pitchers?

Y: ERA

X: Handedness (categorical), xwOBA, K%, HR9

DATA SET: Pitchers statcast metrics from 2025 season

n: 69 pitchers



MODEL SETUP

L: ERA = linear relationship w/ X variables

I: Each pitcher is an independent obs

N: Residuals are normally distributed

E: Residuals have constant variance

$$ERA_i = \beta_0 + \beta_1(xwOBA_i) + \beta_2(K\%_i) + \beta_3(HR/9_i) + \beta_4(Handedness_i) + \epsilon_i$$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.646917	1.850357	-1.971	0.0531
xwOBA	23.529916	5.097672	4.616	1.94e-05

`K%`	-0.006021	0.021927	-0.275	0.7845
HR9	0.411031	0.288877	1.423	0.1596
HandednessRight	-0.048932	0.167497	-0.292	0.7711

Residual standard error: 0.6257 on 64 DF

Multiple R-squared: 0.6786

Adjusted R-squared: 0.6585

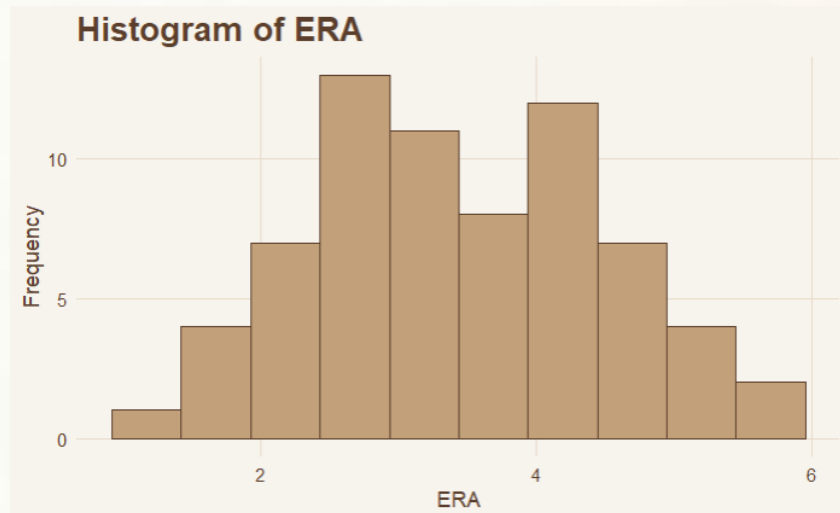
p-value: 3.812e-15

F-statistic: 33.78 on 4 and 64 DF



EXPLANATORY DATA ANALYSIS

The histogram of ERA shows a roughly symmetric, bell-shaped distribution with a slight right skew.



Numerical Variables

Variable	Mean	SD	Min	Max
ERA	3.4289855	1.0708067	1.170	5.70
xwOBA	0.2928116	0.0313178	0.227	0.37
K%	25.8927536	6.3923971	14.200	44.40
HR9	0.9130435	0.3721610	0.200	1.90

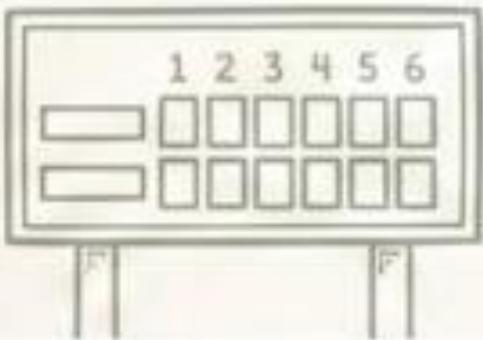
Categorical Variables

Handedness	Proportion
Left	0.3188406
Right	0.6811594



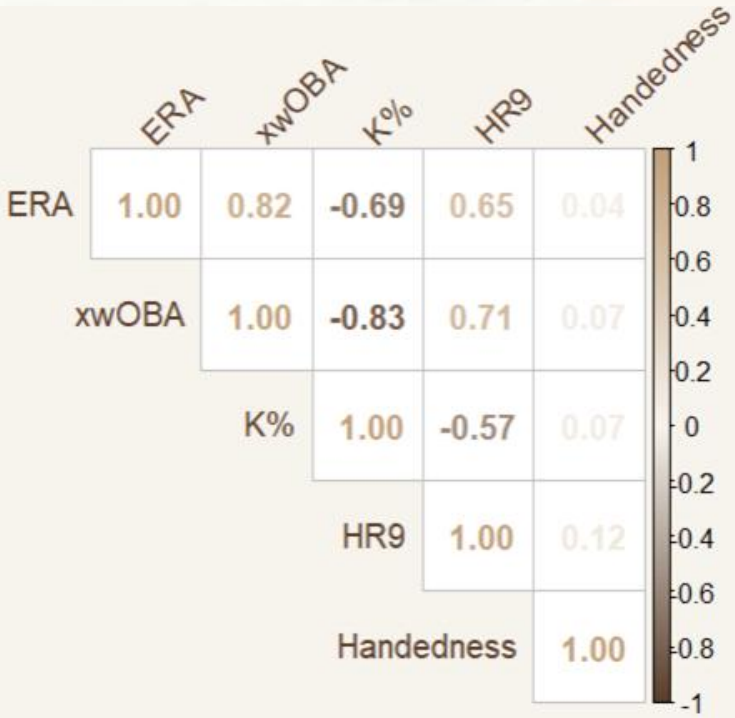
CORRELATION ANALYSIS & MULTICOLLINEARITY PROBLEM

xwOBA and K% are highly correlated, which is reflected in their higher VIF values; however, since all VIFs are below 5, there is no serious multicollinearity issue.



VIF

xwOBA	4.426481
K%	3.412140
HR/9	2.007344
Hand	1.073763





MODEL COMPARISON

The best model is m23 (ERA ~ xwOBA + HR9) because it has the highest Adjusted R² (0.6678).

Model	R2	Adj_R2
m11	0.668	0.663
m12	0.477	0.469
m13	0.418	0.409
m14	0.002	-0.013
m22	0.668	0.658

Model	R2	Adj_R2
m23	0.678	0.668
m24	0.668	0.658
m33	0.678	0.663
m34	0.668	0.653
m4	0.679	0.659



STATISTICAL INFERENCES

xwOBA is statistically significant, with a coefficient of 24.583. A 1-unit increase in xwOBA increases ERA by 24.583; more realistically, a 0.01 increase raises ERA by about 0.246, holding HR/9 constant.

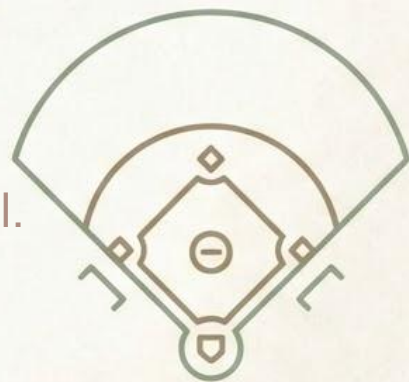
xwOBA ($p < 0.05$): Rejet H_0

HR/9 ($p = 0.163$): Fail to Reject H_0

$$ERA_i = -4.135 + 24.583(xwOBA_i) + 0.401(HR/9_i)$$

term	estimate	std.error	statistic	p.value
(Intercept)	-4.1350765	0.8285486	-4.990747	0.0000046
xwOBA	24.5833161	3.3716569	7.291168	0.0000000
HR9	0.4006185	0.2837286	1.411978	0.1626542

HR/9 is not statistically significant ($p = 0.163$), so we fail to reject the null hypothesis. This means its effect on ERA is not reliably different from zero in this model.

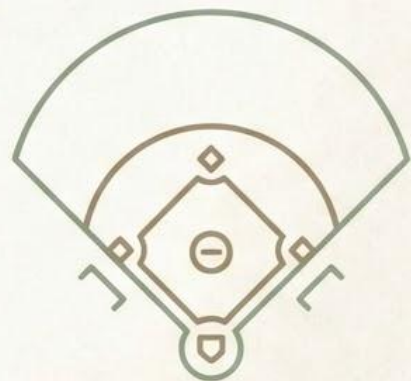


CONFIDENCE INTERVALS

	2.5 %	97.5 %
(Intercept)	-5.7893273	-2.480826
xwOBA	17.8515859	31.315046
HR9	-0.1658641	0.967101

The 95% confidence interval for HR/9 is (-0.166, 0.967), which includes 0, suggesting that HR/9 is not statistically significant and its effect on ERA is uncertain.

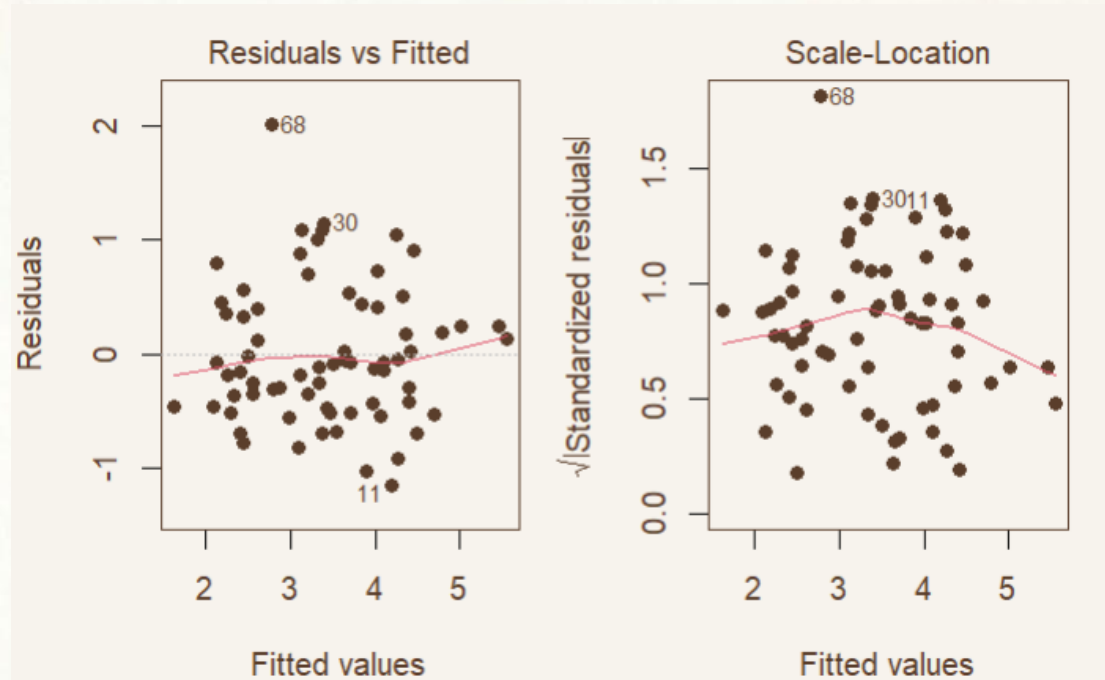
The 95% confidence interval for xwOBA is (17.852, 31.315), which does not include 0, indicating that xwOBA has a statistically significant positive effect on ERA.



DIAGNOSTICS

Assumption: Linearity & Equal Variance (Homoscedasticity)

Observe: Residuals are mostly randomly scattered, but there is a slight curve. Scale-Location plot shows mild variation in spread.

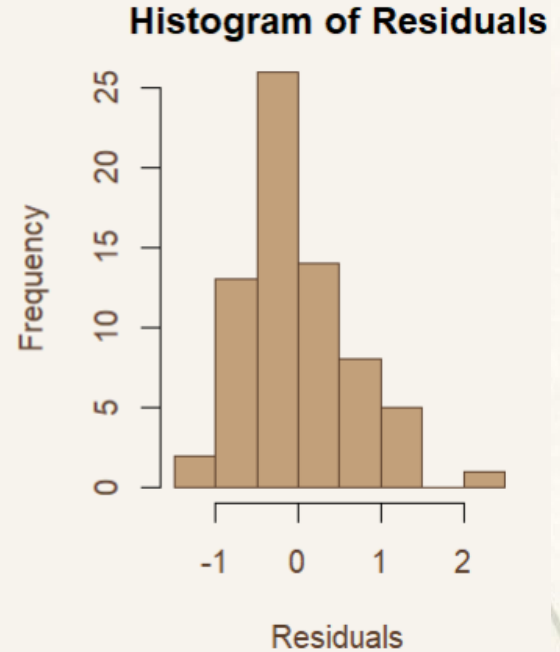
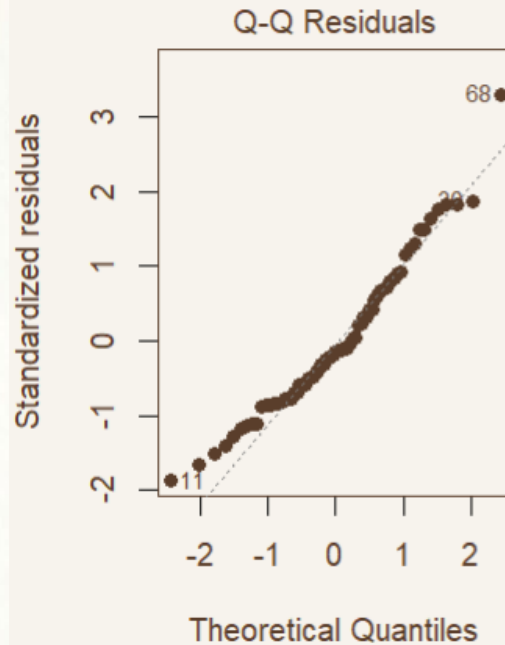


Assumptions partially hold. There is slight evidence of non-linearity and mild heteroscedasticity.

DIAGNOSTICS

Assumption: Normality of residuals

Observe: Points mostly follow the line, but deviate at the tails.
Histogram is roughly bell-shaped with slight skew.

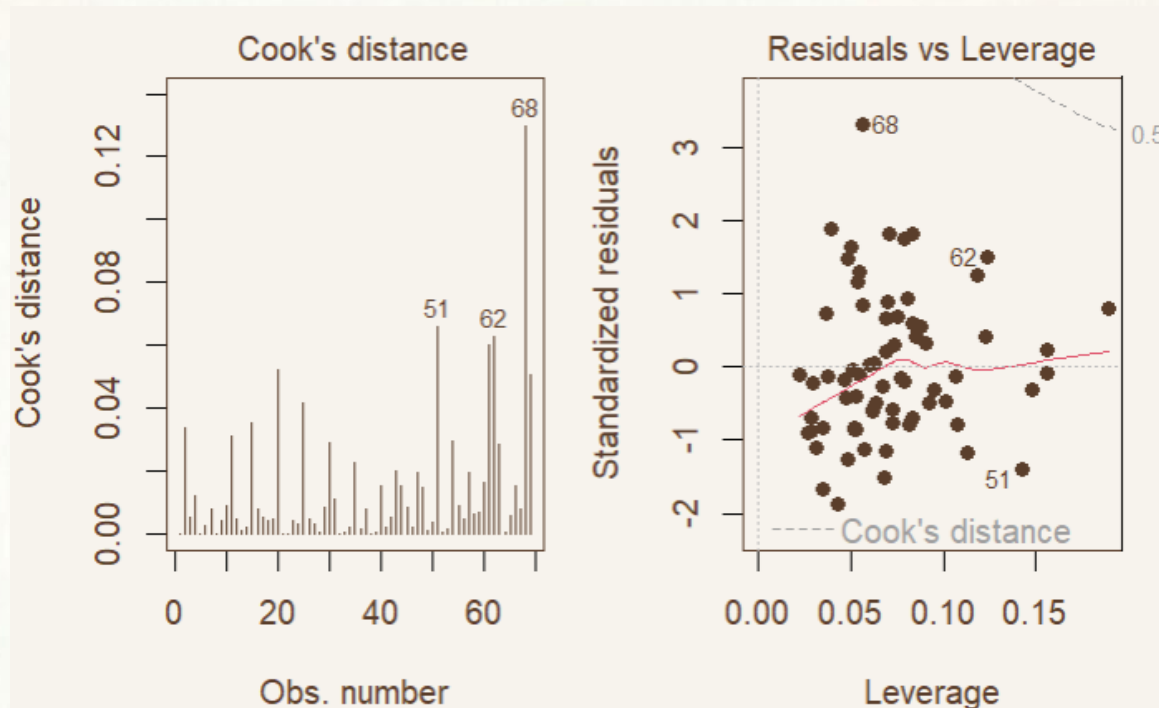


Normality assumption partially holds.

DIAGNOSTICS

Assumption: Influential observations

Observe: A few points (especially 68, also 51 and 62) have higher Cook's distance and leverage.



There are some influential observations, but none appear extreme.



CONCLUSION

Summary of Key Findings

The model reveals that ERA is most strongly related to xwOBA, with weaker relationships from K%, HR/9, and handedness. xwOBA was the most important variable in the model.

Statistical Significance

xwOBA was the only statistically significant variable in the model. While the other were not, this implies their effects are not reliably different from zero in this model.

Practical Interpretation

These results mean that allowing higher-quality contact leads to higher ERA in real-world pitching performance. The model can be used to evaluate pitchers and guide development by focusing on reducing contact quality.

